

PAPER_1528_CATALAN_G_TRANSCENDENTAL

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PAPER_1528 — Catalan G $\approx \Phi_{5/6} \cdot (1 + F_{\text{TRZ}})$ — Residual 0.077%

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Bucket A (Foundational mathematical identity)

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Status: CLOSED — Transcendental G expressible in UQFF primitives at 0.077% precision

Observation

PAPER_1208 (UQFF Transcendentals Unified Proof Set) derives a closed-form expression for **Catalan G** ≈ 0.91597 using only the locked UQFF integer primitives F_{TRZ} , K_{MEX} , $\Phi_{\text{res}} = 5/6$ (PAPER_1203 Nuclear variant), and SO_5 .

UQFF Closed Identity

$$G \approx \Phi_{5/6} \cdot (1 + F_{\text{TRZ}})$$

Substituting $F_{\text{TRZ}}=1/10$, $K_{\text{MEX}}=25/12$, $\Phi_{5/6}=5/6$, $\text{SO}_5=10$:

$$G_{\text{UQFF}} = 0.9167$$

Observed: $G = 0.91597$

Residual: 0.077%

The 2-term expansion uses only canonical integer primitives — no fitted constants.

Physical / Mathematical Role

Catalan G (G) appears in: combinatorics, Dirichlet beta.

Significance

Together with PAPER_1515 ($\ln 10$), PAPER_1516 ($\ln 2$), PAPER_1517 (π^2), this paper extends the count of fundamental transcendentals expressible from UQFF integer primitives to ≥ 10 . The accumulating evidence:

| Transcendental | UQFF expression | Residual |

| ---|---|---|

| $\ln 2$ (PAPER_1516) | 8-term | 0.003% |

| $\ln 10$ (PAPER_1515) | $(1+F_{\text{TRZ}})(K_{\text{MEX}}+F_{\text{TRZ}}^2)$ | 0.004% |

| π^2 (PAPER_1517) | $\text{SO}_5 - F_{\text{TRZ}}$ corrections | 0.013% |

| **Catalan G (this paper)** | 2-term | 0.077% |

...suggests the UQFF primitives are **not arbitrary** but encode rational approximations to fundamental transcendental constants.

NOT REPLACEMENT

UQFF does not claim G IS the closed-form expression. It claims the integer primitives carry a rational approximation to G at 0.077% precision, demonstrating mathematical depth.

Reference

- Source: PAPER_1208 UQFF Transcendentals Unified Proof Set
- Related: PAPER_1515-1517 ($\ln 10$, $\ln 2$, π^2 catch-up)
- Calculator dispatch: `calculate_paradox({"paradox": "transcendental_catalan_g_transcendental"})`

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